– they make it more likely that an organisms will survive and reproduce successfully in that environment. This is known as **convergent evolution**. An example is the wing of a bat and the wing of an insect (Figure **D.13**). The wings serve similar functions but they are derived from completely different structures. Bats and insects are not closely related.

Figure **D.14** summarises the differences between convergent and divergent evolution.

Figure D.13 The bat's wing is structurally very different from the insect's wing, yet they perform a similar function. This is an example of convergent evolution.

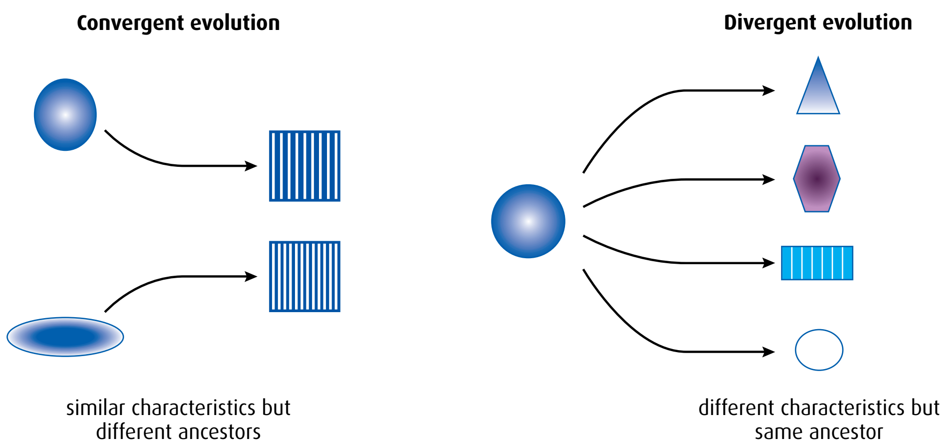

Figure D.14 These diagrams show the difference between convergent and divergent evolution.

The pace of evolution: gradualism and punctuated equilibrium

Darwin viewed evolution as a slow, steady process called **gradualism**, whereby changes slowly accumulated over many generations and led to speciation. For many species, this seems to be true. A good example of gradualism is the evolution of the horse's limbs, which fossils indicate took around 43 million years to change from the ancestral form to the modern one (Figure **D.15**, overleaf).

In some cases, the fossil record does not contain any intermediate stages between one species and another. A suggested explanation is that fossilisation is such a rare event that the intermediate fossils have simply not been discovered. In 1972, Stephen J Gould (1941–2002) and Niles Eldredge (b. 1943) suggested that the fossils had not been found because

Organism	Number of amino acid differences from human
human	0
chimpanzee	0
rhesus monkey	1
rabbit	9
mouse	9

Table D.5 Variation in the amino acid sequence of cytochrome c.

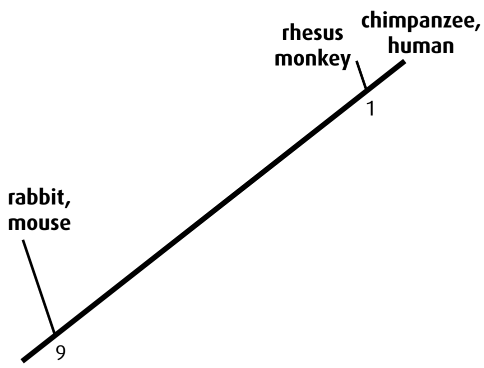

Figure D.27 A cladogram for five mammal species.

Cladograms and the classification of living organisms

Cladograms or phylogenetic trees can be produced using information from fossils, morphology, physiology and behaviour, or molecular data. In most cases, they will agree with each other, or the differences will be small. When there is uncertainty, a second cladogram, built up using a different feature can be constructed for comparison. Each cladogram can be thought of as a hypothesis about the relationships of the organisms it contains.

A disadvantage of the cladogram shown in Figure **D.26** is that gives no indication of how close or far apart the branching points are. But if molecular data are used then the cladogram can be drawn to scale based on the number of molecular differences. The cytochrome c differences given on page **390** can be tabulated as shown in Table **D.5**.

There are no differences between rabbit and mouse so they have to be drawn together at the end of a branch and the same applies to the chimpanzee and human. Rhesus monkey differs from chimpanzee and human by only one amino acid and so the branch point must be 1 unit from the end. Rabbit and mouse differ by 9 amino acids and so the branch point must be 9 units further down.

The cladogram could be simply drawn as shown in Figure **D.27**.

Biochemical analysis of other molecules or comparison of DNA sequences would be needed to complete the separation of rabbit from mouse and human from chimpanzee.

Occasionally a cladogram based on molecular data indicates an evolutionary relationship that disagrees with other or older versions. There are two possible reasons for this. First, convergent evolution can bring about morphological similarities that could be misleading, and secondly the molecular interpretation is based on the probability of change occurring at the same rate in a 'Darwinian' manner. This may not be the case as chance events could cause it to speed up or slow down and DNA analysis can be obscured by multiple point mutations which make it difficult to identify the 'true' evolution of a stretch of DNA. Nevertheless, where the different methods support each other, the evidence is more robust and where there is disagreement, it provokes scientific debate and the search for further evidence to support either one argument or the other. Cladistics is not a subjective method of classification as older methods used to be and as such it is a good example of the scientific method of studying organisms.

13 Explain the difference between homologous and analogous characteristics.

14 Outline how biochemical differences can be used to give an indication of the pace of evolution.

15 Outline the reasons why it is useful to classify organisms.